Retail Enterprise Database System

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Course: CSC 411 Database Management Systems Design H002

Professor: Dr. Bo Li

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1. Abstract

The following project focused on designing and implementing a retail enterprise database system using MySQL and then integrating it with a Java-based web application.

This system is responsible for managing data related to stores, products, brands, and sales transactions. A simple yet effective web interface was developed in order to execute analytical queries and then display results dynamically.

This project demonstrates a proper use of SQL, relational database design, and Java JDBC to come up with a working database-driven application.

2. Problem Description

Retail businesses tend to generate large amounts of data such as product sales, store performance, and purchasing patterns. If there is a lack of a structured system it becomes very difficult to manage and analyze this data.

The primary goal of this project was to design a system that had the following features:

- It stores retail data in an organized way.
- It maintains relationships between several entities.
- It supports analytical queries and helps with business insights.

This system includes a simple yet user-friendly web interface for users to run queries and view the results easily.

3. Database Design

Here I designed the database using multiple related tables in order to represent an actual retail enterprise system.

Main Tables

Here is the list of the main tables:

- retail_store
- product
- brand
- product_type
- sale
- sale_item

Each of these tables have a primary key, and the relationships are defined using foreign keys.

Relationships:

Here is the list of the relationships:

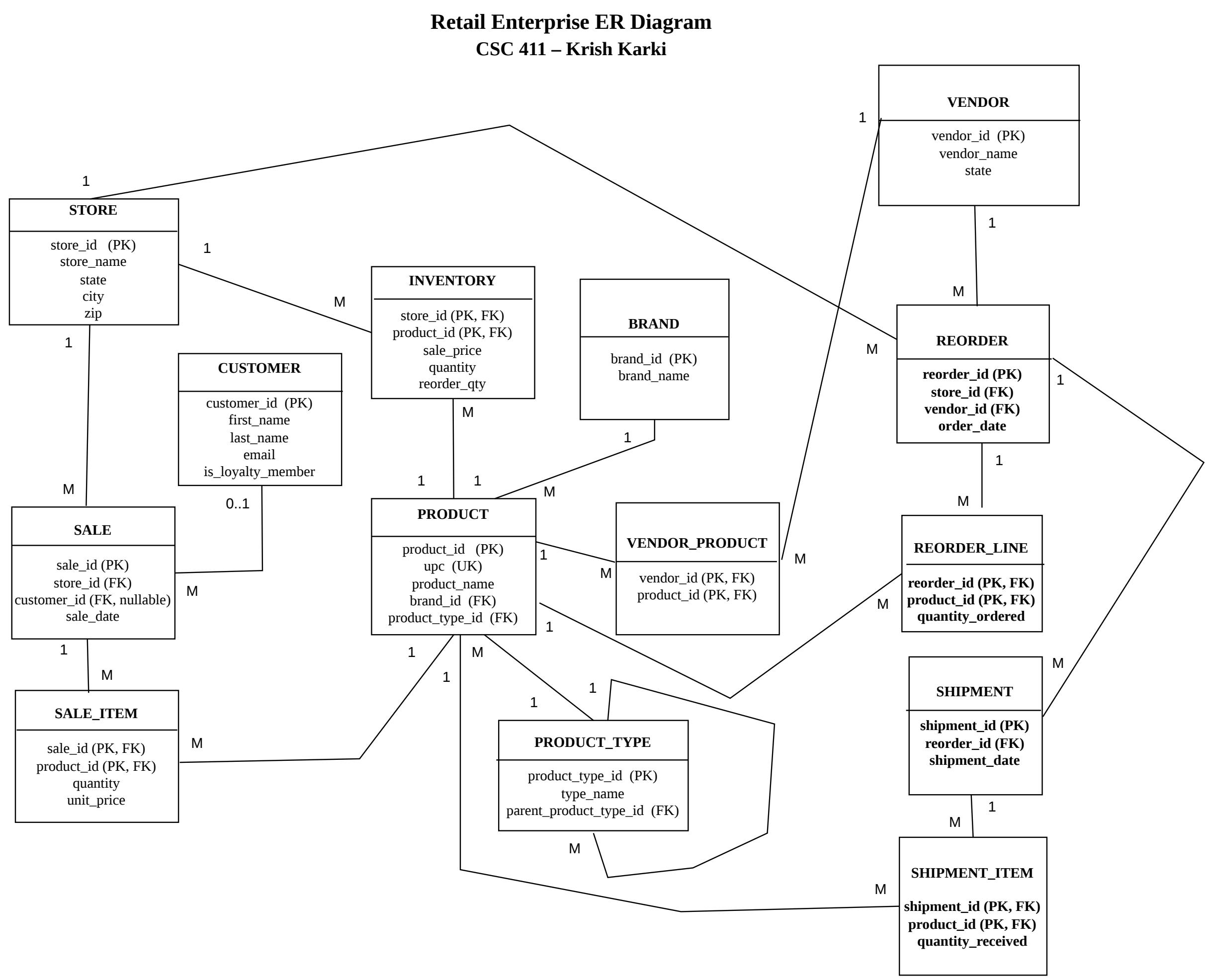
- A store is associated with multiple sales.
- A sale contains multiple sale items.
- Each product belongs to a brand and a product type.

This design keeps the data organized and avoids any possible duplication.

Relational Schema:

I implemented the relational schema in the file 01_schema.sql which defines all tables, primary keys, and foreign key relationships.

E-R Diagram



Note:
- INVENTORY, SALE_ITEM, VENDOR_PRODUCT, REORDER_LINE, and SHIPMENT_ITEM use composite primary keys
- SALE.customer_id is nullable to support anonymous transactions

4. Implementation Details

4.1 Database Implementation

I implemented the database using MySQL with the following SQL files:

- 01_schema.sql creates tables and relationships.
- 02_sample_data.sql inserts sample data.
- 03_required_queries.sql contains required analytical queries.

4.2 Java Web Application

I developed a Java application in order to connect it to the database using JDBC and execute the necessary queries.

Here are the main components:

- DatabaseConfig.java loads database configuration.
- QueryService.java executes queries and formats results.
- RequiredQueries.java stores SQL queries.
- RetailWebServer.java runs the local web server.

As the application uses Java's built-in HTTP servers no external server is required.

4.3 User Interface

I developed a simple yet user-friendly interface using HTML, CSS, and JavaScript.

Here are the features of the user-interface:

- It has buttons for each query.

- It does dynamic loading of results.

- It gives table-based output

Using the interface users can run queries without directly having to write any SQL.

5. Running Results and Analysis

I was successful in executing all five required queries using the buttons in the web application.

Results:

- Query outputs are displayed in table format in the browser.
- Each query returns correct results based on the sample data.

The screenshots below show the web interface and the successful execution of queries with actual data output.

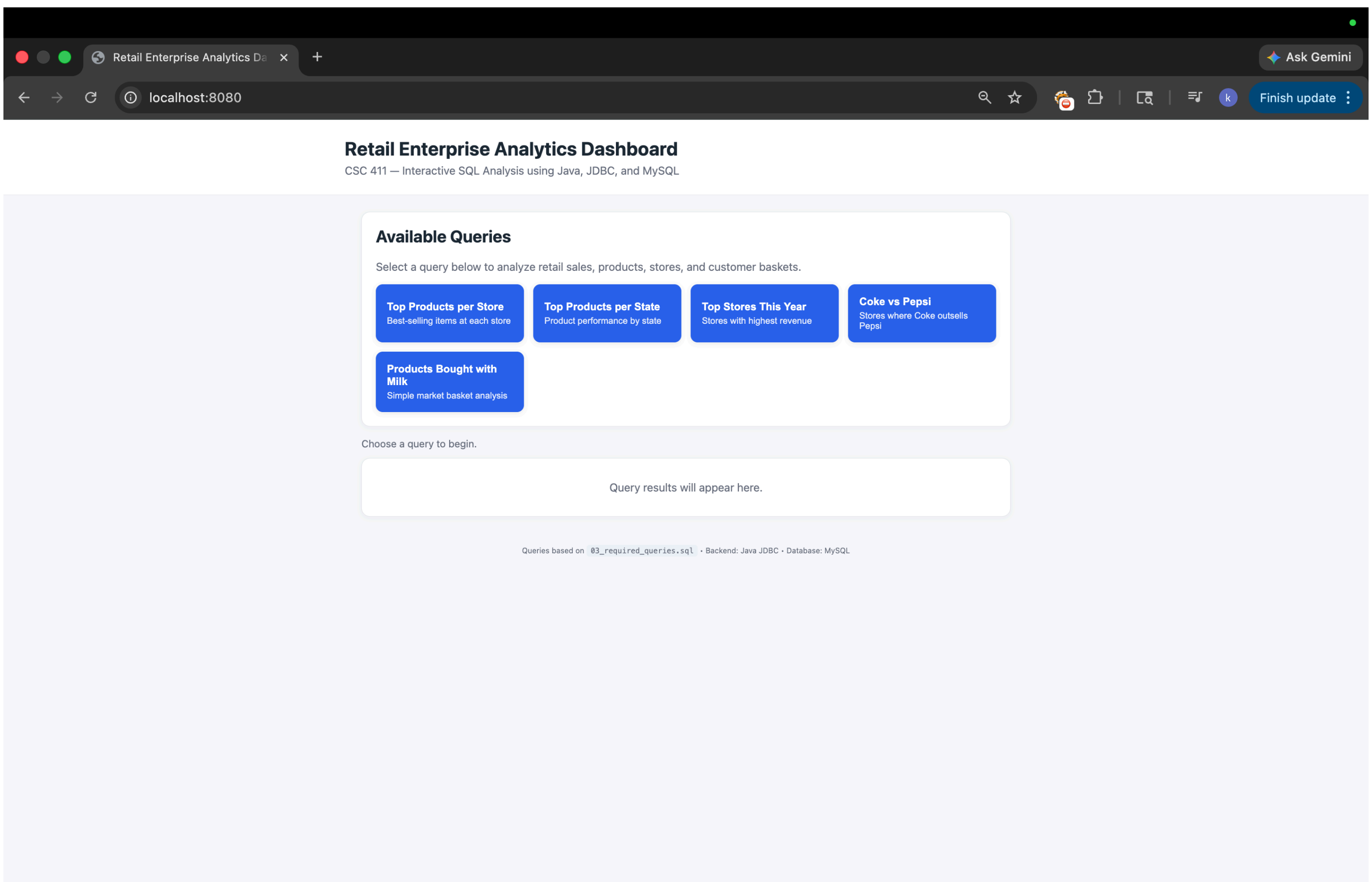


Figure 1: Retail Enterprise Web Application Homepage

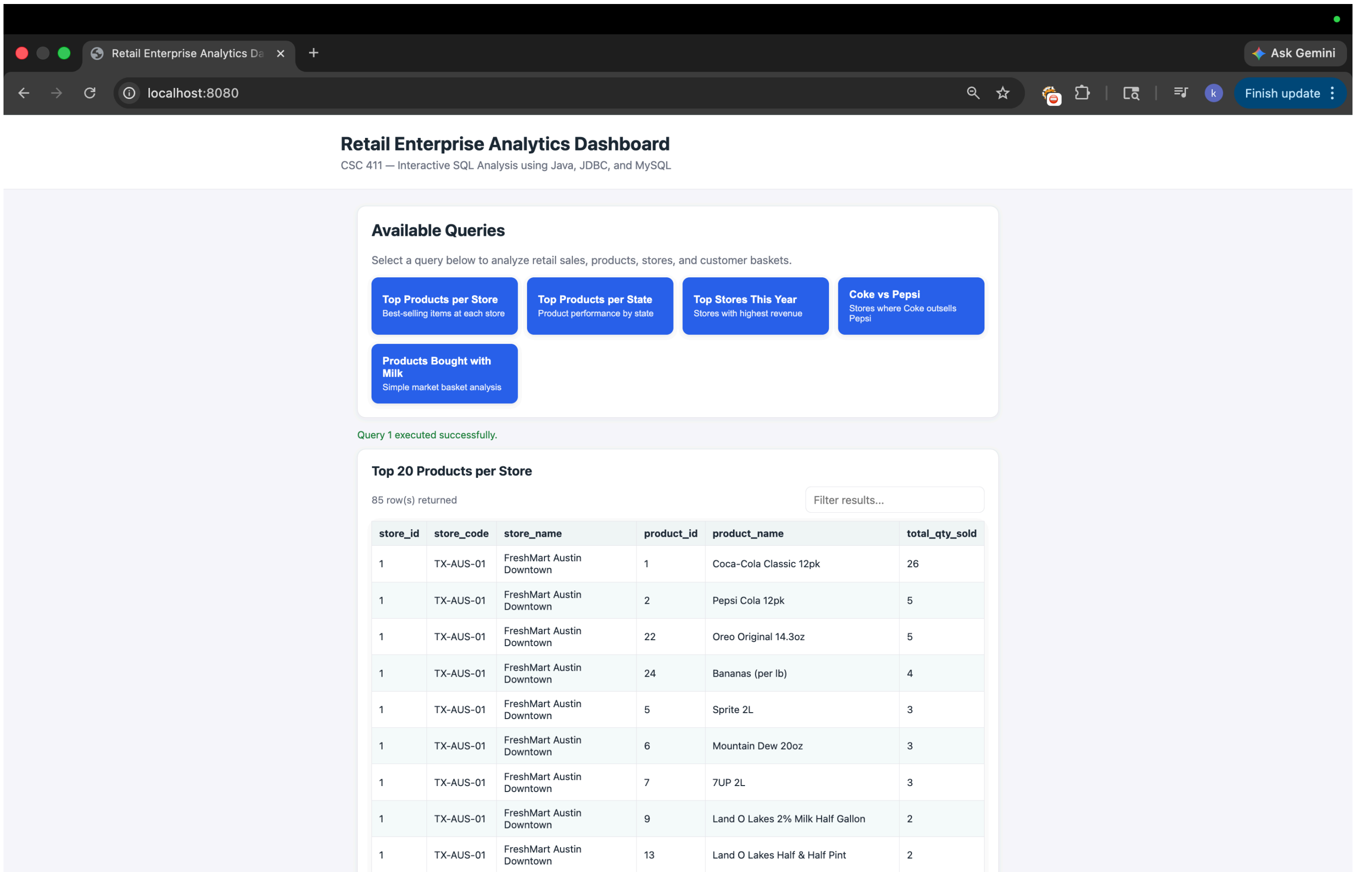


Figure 2: Sample output of Query 1 (Top Products per Store)

Analysis:

- Query 1 and Query 2 identify best-performing products
- Query 3 shows store performance based on total sales
- Query 4 compares Coca-Cola and Pepsi sales across stores
- Query 5 shows products frequently purchased together with milk

These queries demonstrate it very well how the database can be used for business analysis.

6. Conclusion

My project tends to demonstrate the design and implementation of a relational database system that is integrated with a Java-based application.

Through the entirety of this project I was able to learn the following:

- I learned how to design relational databases.
- I learned how to write and execute SQL queries.
- I learned how to connect a database using Java JDBC
- I learned how to display results through a web interface.

As a conclusion, this project provided me with a practical experience in building a complete database-driven system.

7. References

I did not use any external references. All of the work is based on course materials and project requirements provided to us by Dr. Li.